RAKESH BORRA

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OBJECTIVE

MS Computer Science student with experience in full-stack application development, backend API integration, database-driven applications, and Python-based workflows, seeking entry-level Software Engineer and Backend Developer opportunities.

EDUCATION

University of Massachusetts Boston, MA, USA **Jan 2025 – Present**
Master of Science in Computer Science

Indian Institute of Information Technology Sricity, India
Bachelor of Technology in Computer Science and Engineering

SKILLS

- **Programming Languages:** Python, JavaScript, SQL, Java
- **Backend Development:** Node.js, Express.js, FastAPI, REST APIs
- **Frontend & Databases:** React, HTML, CSS, PostgreSQL, MongoDB, Firestore, SQLite
- **Tools & Concepts:** Git, GitHub, Docker, pytest, Data Structures, OOP

EXPERIENCE

Graduate Teaching Assistant – PHYS 113 **Sep 2026 – Present**
University of Massachusetts Boston

- Evaluate quantitative assignments and exams, identify errors, and provide consistent, detailed feedback.
- Explain technical problem-solving methods during office hours and coordinate grading with instructional staff.

Dining Services Associate **Jan 2026 – Aug 2026**
Sodexo – University of Massachusetts Boston

- Collaborated with team members to support time-sensitive operations and provide responsive service in a fast-paced environment.

PROJECTS

Real-Time Smart Traffic Monitoring and Congestion Prediction System (CV + Full Stack)

- Built a traffic monitoring platform using YOLO, OpenCV, FastAPI, and React for vehicle detection, lane-wise counting, and congestion analysis from live video streams.
- Developed real-time dashboards, traffic density prediction, PostgreSQL integration, and Docker-based deployment.

Real-Time Real Estate Investment Analysis Web Application (React / Firebase)

- Developed a full-stack application using React, Vite, Firebase Authentication, and Firestore for authenticated user workflows.
- Integrated property-data APIs and implemented ROI, cash flow, and cap rate calculations with five-year projections and user-scoped database access rules.

Clinical Trial AI – Protocol Analysis Platform (Python / Streamlit)

- Collaborated in a team of three on a Python application to extract and structure clinical trial PDF information using PyMuPDF, OCR, and locally hosted language models.
- Developed source-page verification and automated testing workflows to support traceable summaries and identify missing data.

Filter Pruning in Deep Neural Networks (Machine Learning)

- Applied structured filter pruning to convolutional neural networks to reduce inference latency while preserving accuracy.
- Achieved up to 2x speedup through statistical analysis and evaluation of pruning strategies.

CERTIFICATIONS

The Complete Data Structures and Algorithms Course in Python – Udemy | View Certificate